MASTER'S THESIS DEFENSE · BIOMEDICAL ENGINEERING

Development of a Clinical Liver MRI Dataset for Liver Tumor Detection Using Deep Learning

A phase-aware, provenance-documented dataset and a controlled arterial-only versus four-phase nnU-Net experiment

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OUTLINE

What this presentation covers

01

Background and problem

Liver cancer burden, multiphase MRI and the data bottleneck

02

Gap and research questions

What is missing in public liver MRI data, and four questions

03

Literature review

Architectures, datasets and the tumor-versus-liver Dice gap

04

Methodology

Cohort, preprocessing, patient-disjoint splits, A-versus-B design

05

Results

Dataset profile, detection, segmentation and uncertainty

06

Conclusion

Contributions, limitations and future work

BACKGROUND

Liver cancer: a major global burden

3rd

leading cause of cancer-related death worldwide; among the six most diagnosed cancers [1]–[3]

~80%

of primary liver malignancies are hepatocellular carcinoma (HCC) [4]

~18%

overall five-year survival, driven mainly by stage at presentation [5]

Most patients present at intermediate or advanced stages, when resection, transplantation or ablation is no longer possible — so **early, accurate detection decides access to curative therapy** [1], [2], [6].

BACKGROUND

Multiphase MRI: rich information, hard to use

Why MRI

- Reference non-invasive modality for focal liver lesions, superior soft-tissue contrast, no ionizing radiation [3], [7]–[9]
- Dynamic contrast phases — native, arterial, portal venous, delayed (and hepatobiliary) — each show a different aspect of perfusion [2], [6], [10]
- Diagnosis depends on reading the full series together, not one image [10], [11]

Why it is hard

- No fixed intensity scale: gray values change with vendor, field strength and sequence [7], [8], [15]
- Breath-holds misalign the phases [5]–[7], [9]
- Manual 3D delineation takes hours per patient [12], [13]
- Expert lesion Dice ≈ 0.74–0.78 versus > 0.95 for whole liver [13], [15]

PROBLEM STATEMENT

Four challenges in turning clinical MRI into a detector

Multiphase DICOM integration

Combine several 3D contrast phases without losing perfusion information or distorting space [3], [7], [8]

Anatomical and boundary variability

HCC varies in size, shape and signal, often inside cirrhotic or fibrotic livers [3], [6], [15]

Spatial class imbalance

The tumor fills a tiny fraction of the volume, biasing losses toward background [3], [12], [21]

Small-lesion detection

Small, early lesions matter most but have the fewest voxels and weakest contrast [12], [15]

Premise of the thesis: reliable automated detection is today **more a data-engineering problem than an architecture problem** [3], [8], [15].

RESEARCH GAP

No standardized, leakage-free liver MRI dataset exists

CT

Public benchmarks are CT-dominated (LiTS); MRI collections are small, single-centre and rarely publish splits or provenance [7], [8], [17], [22]

29%

average VIDS compliance of public datasets; provenance only 8%, quality documentation 25% [22]

0.67 → 0.21

foreground Dice after enforcing patient-disjoint splits — slice-mixed splits inflate results [24]

Missing: a **phase-aware, provenance-documented clinical liver MRI dataset** that is machine-learning-ready and evaluated without leakage [7], [8], [22], [23].

Four questions, five objectives

RQ1 Can a clinical single-centre multiphase corpus be standardized into a traceable, phase-aware, ML-ready dataset?

RQ2 How do lesion size, class balance, visibility and annotation coverage constrain what a detector can learn?

RQ3 Does a four-phase input improve HCC detection and segmentation over arterial-only, under an identical protocol?

RQ4 How uncertain is the multiphase advantage, and what sample size would establish it?

Objectives

1. Acquisition and curation requirements [6], [7]
2. Reproducible annotation and provenance protocol [6], [13], [22]
3. Preprocessing and input-representation pipeline [7], [15], [25]
4. nnU-Net 2D benchmark: arterial-only vs four-phase [3], [7], [12]
5. Evaluation and reporting standards [22]–[24]

LITERATURE REVIEW

Liver is nearly solved; tumors are not

Model (family)	Data	Liver Dice	Tumor Dice	Ref
Attention U-Net++	ATLAS, 60 cases	0.94	0.85	[36]
3D U-Net + ensemble	Late-phase MRI, 107 patients	—	0.74	[15]
nnU-Net vs Swin UNETR	Multiphase T1-VIBE, 458 scans	0.97 / 0.95	0.56 / 0.42	[7]
nnU-Net (hepatobiliary)	HBP MRI, surgical planning	0.97	0.77	[12]
SWTR-Unet (CNN + transformer)	1.5 T MRI, 48 cases	0.98	0.81	[3]
Cross-modality 3D U-Net	CirrMRI600+	0.94	—	[32]

Liver Dice clusters at **0.93–0.98**; tumor Dice spans **0.44–0.85** with phase, dataset and label definition. Human inter-rater tumor Dice is only **0.78** [15] — part of the “error” is label noise.

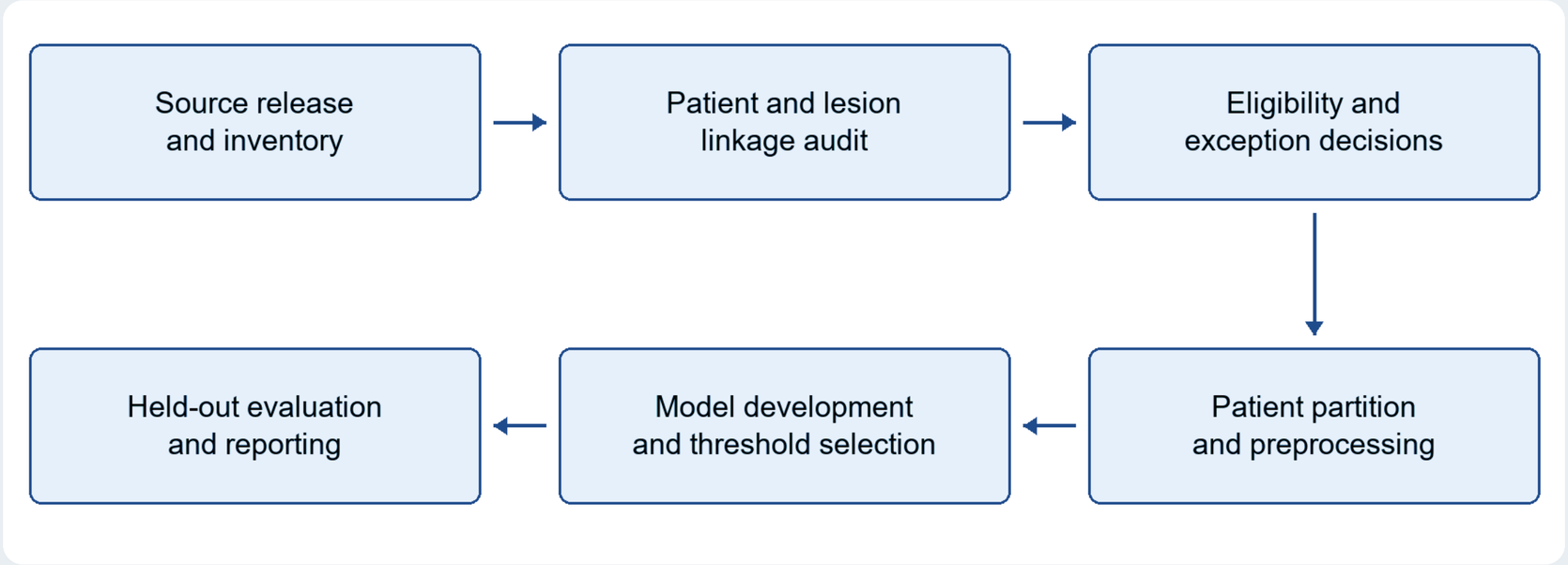
Public liver MRI datasets remain few and narrow

Dataset	Patients / scans	Sequences / phases	Annotations	Ref
OpenSwissHCC / ATLAS	60 (+30 private)	CE T1, 3 phases	Liver + HCC	[36], [37]
LiverHccSeg	17	T1 arterial	2-rater liver + HCC	[13], [38]
Duke Liver	105 / 2,146 series	Multi-sequence	Whole-liver masks	[8], [39]
CirrMRI600+	339 / 628 scans	T1 venous + T2	Cirrhotic liver	[8]
LiQA	440–610	T1/T2/DWI + 4 phases	Liver + fibrosis stage	[9]
LiTS (CT reference)	201 CT scans	CE-CT, 1 phase	Liver + tumor	[17]

Reported performance is partly a property of the dataset. This thesis adopts published conventions: **BIDS-like folders, NIfTI + JSON sidecars, per-annotation provenance and a split manifest**.

METHODOLOGY

A design-based, fully reproducible pipeline



Seven ordered stages — **audit, prepare, freeze, develop, calibrate, finalize, evaluate** — each writes a frozen, hashed artifact consumed by the next. Only the input representation differs between experiments.

METHODOLOGY

Cohort construction: 44 → 38 → 30 / 8 patients

44

Source subjects, all four dynamic phases; all pass the arterial-reference check

→

38

Retained after technical QC: 33 HCC-positive, 5 negative

→

30 development patients

26 positive / 4 negative · 38 lesions

8 held-out test patients

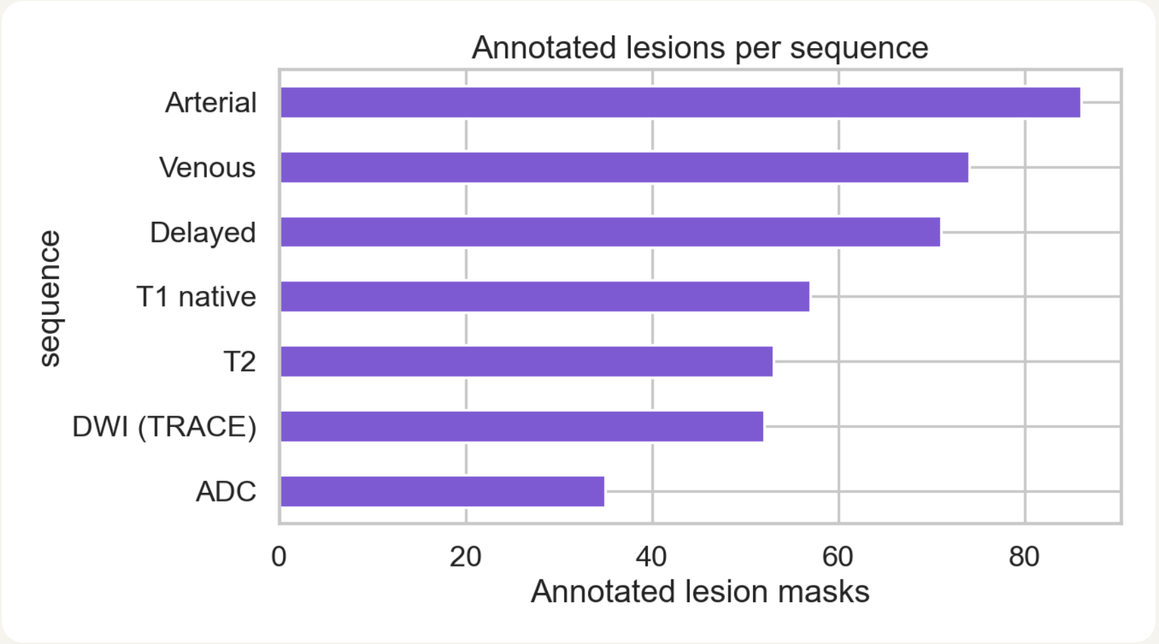
7 positive / 1 negative · 15 lesions

Source: Lausanne University Hospital, retrospective, 2011–2020; Skyra scanner for 72 of 89 lesion rows.

6 exclusions (one objective rule): the lesion mask was drawn on a different arterial acquisition than the highest-TTC one selected.

RESULTS · DATASET

A small-lesion, arterial-visible, mixed-truth dataset



89
annotated lesions in 39 positive subjects

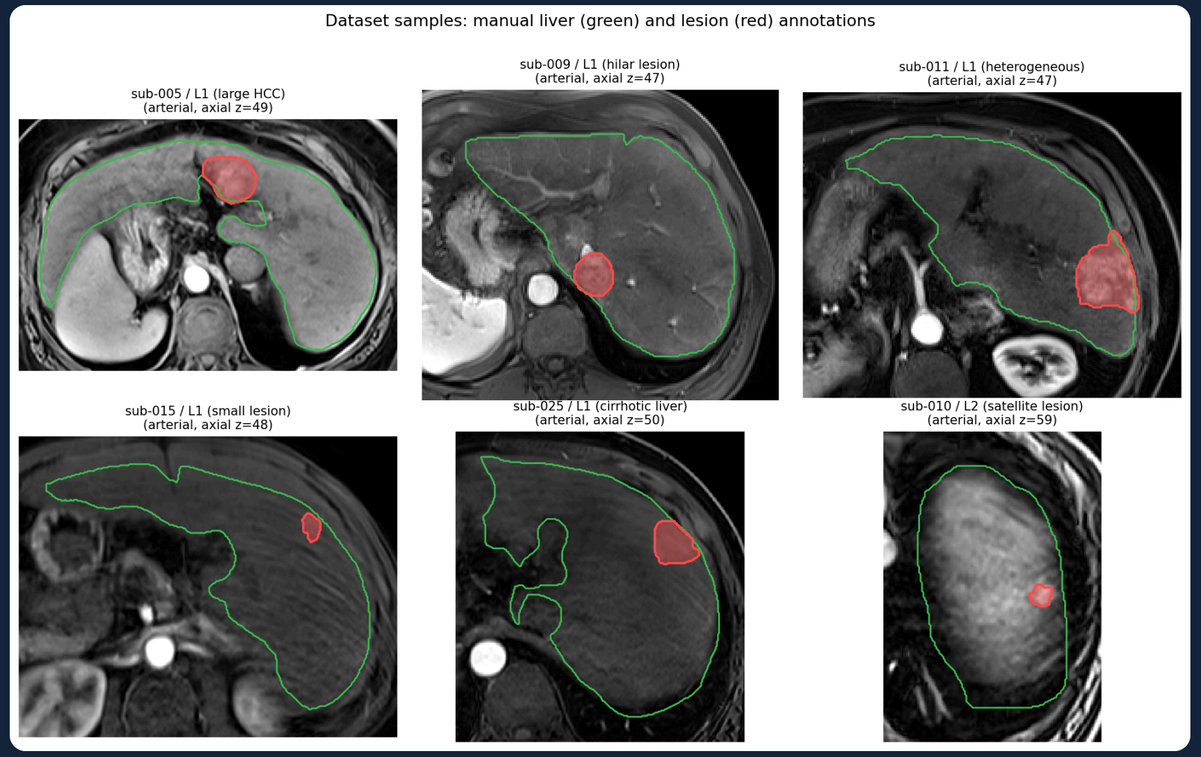
55/89
lesions under 20 mm (median 17 mm)

96.6%
visible on the arterial phase (ADC 36.0%)

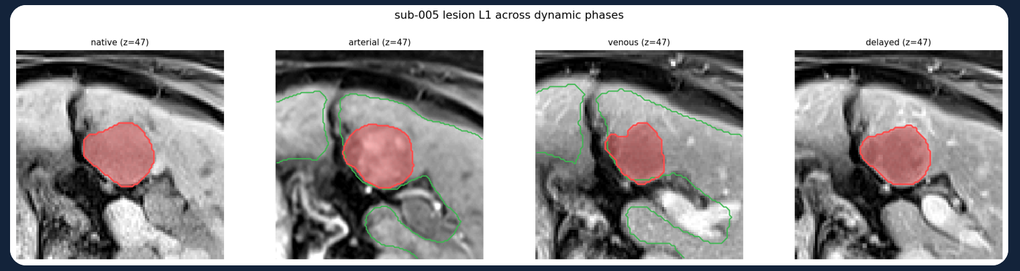
24/89
biopsy-proven; 38 LI-RADS 5, 27 unrecorded

Release: 766 NIfTI volumes, 427 manual lesion masks, 91 manual and 236 automated liver masks, 516 registration transforms; 43 of 47 metadata columns complete.

What the annotated data look like



Six lesions: liver contour (green), lesion mask (red)



One lesion across native, arterial, venous and delayed phases — the four-channel input of Experiment B

Lesions vary in size, location and contrast; each phase shows the same lesion differently, which is the information a multiphase model can exploit.

METHODOLOGY

One variable changes: the input phases

Experiment A — arterial only

1 channel: the arterial reference volume

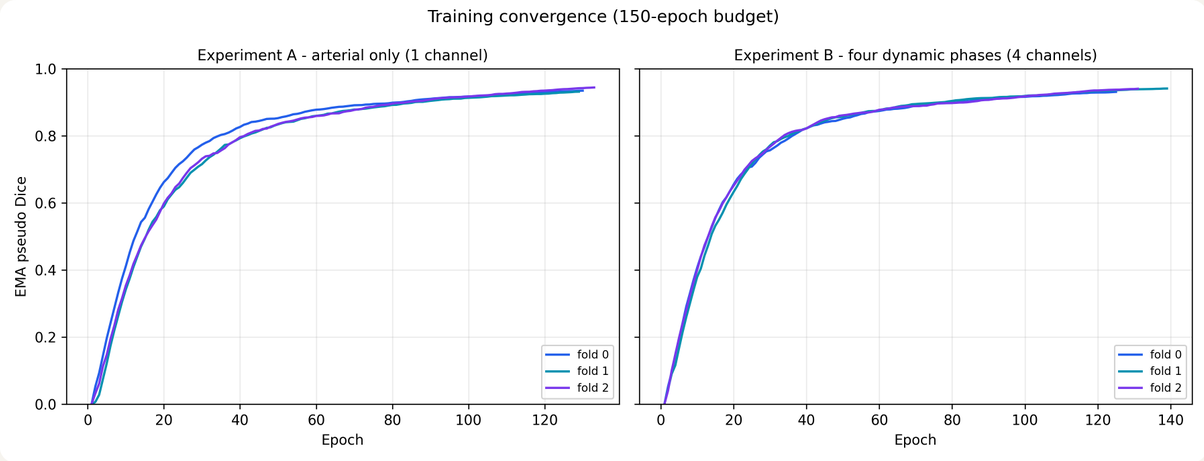
Experiment B — four phases

4 channels: native, arterial, venous, delayed

Held fixed in both arms	Setting
Preprocessing	Per-case foreground z-score; rigid registration (coverage ≥ 0.880)
Splits	Patient-disjoint 30 / 8; 3 class-stratified folds; seed 2026
Model and training	nnU-Net v2.6.2, 2D; 150 epochs; SGD, Dice + CE loss; EMA weights
Operating point	Max OOF lesion sensitivity at ≤ 1 FP per patient; frozen before test
Evaluation	Cardinality-first matching at IoU ≥ 0.10; 2,000 patient bootstraps

RESULTS · DEVELOPMENT

Training converged; out-of-fold, B finds twice as many lesions



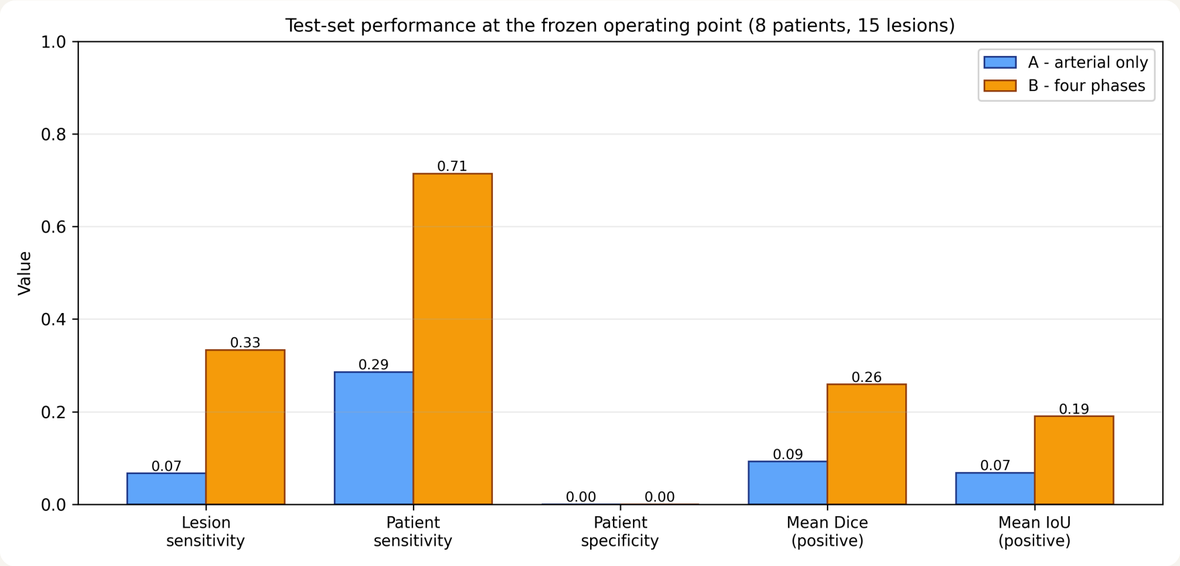
Final EMA pseudo-Dice 0.930–0.950 in all six jobs; curves flatten after epoch ~100–120.

Out-of-fold (30 patients)	A	B
Threshold	0.83	0.89
Matched lesions	6 / 38	13 / 38
Lesion sensitivity	0.158	0.342
FP per patient	0.567	0.733
Patient sensitivity	0.500	0.692
Mean Dice (positive)	0.100	0.220
ROC AUC	0.654	0.538

Experiment B was selected by the pre-set rule: it detected more than twice as many lesions under the false-positive budget.

RESULTS · HELD-OUT TEST SET (8 PATIENTS, 15 LESIONS)

Four phases detect five times more lesions



Lesions found

1 → 5

of 15 (A → B)

Mean Dice

0.092 → 0.259

2.8× higher

Patient sensitivity

0.29 → 0.71

PPV 0.67 → 0.83

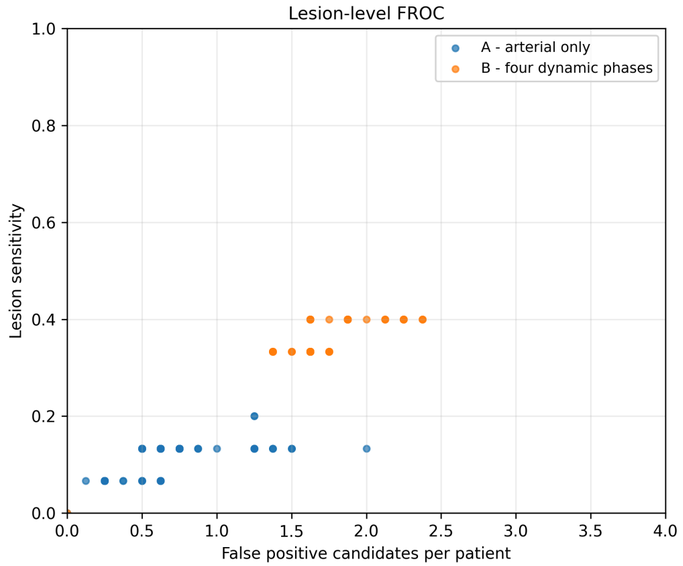
FP per patient

0.25 → 1.50

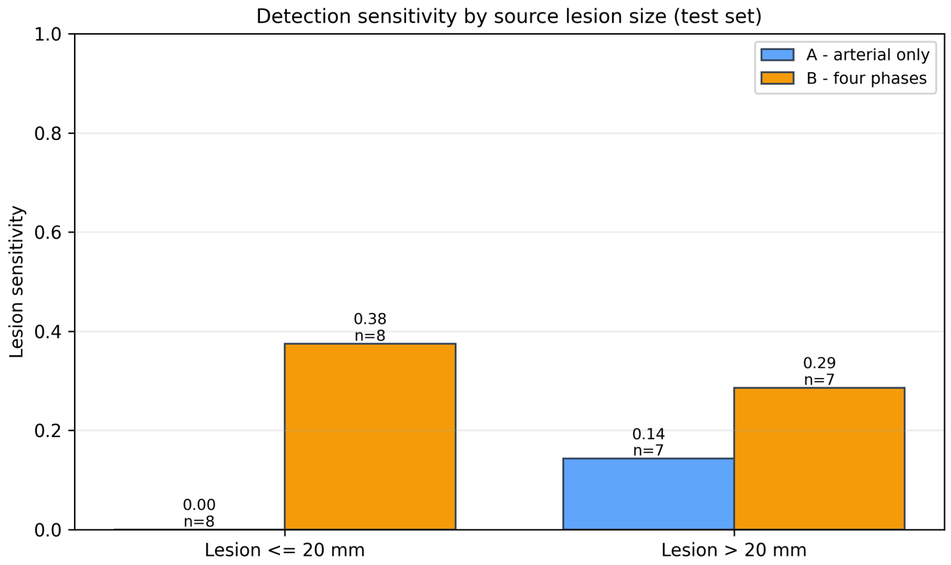
the price of sensitivity

Specificity, NPV and ROC AUC rest on a single negative test patient (n = 1) and are not interpretable.

The gain holds across operating points and sits in small lesions



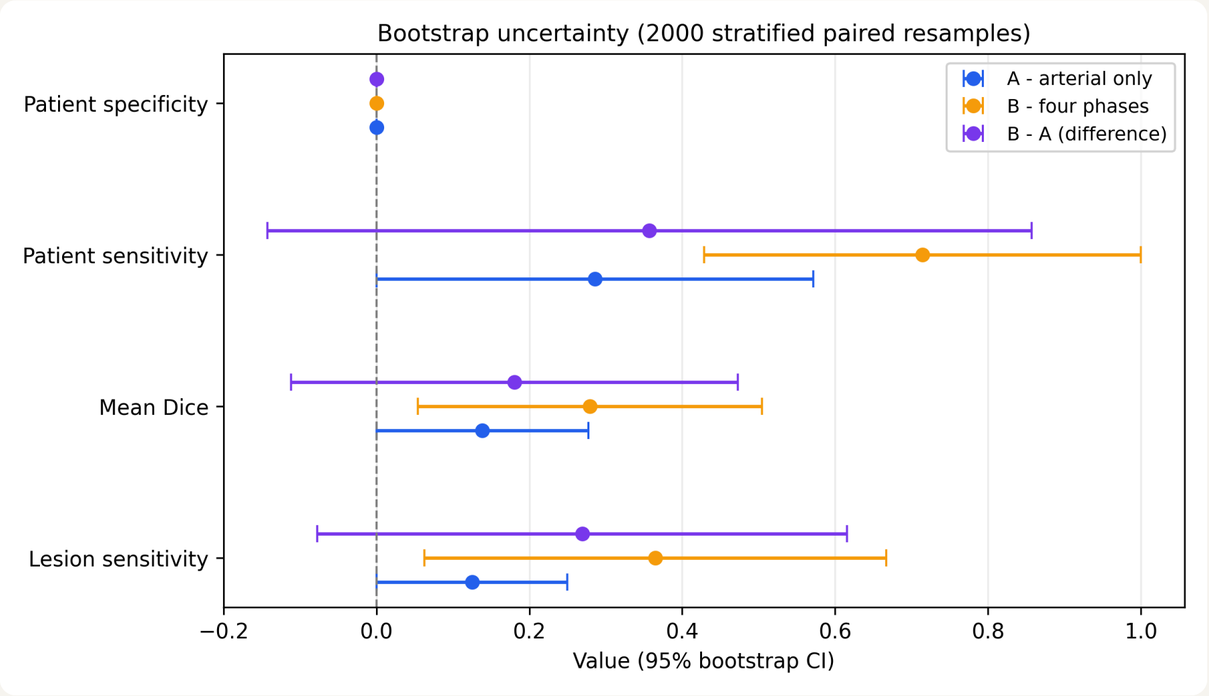
FROC: B lies above A everywhere; at 0.7 FP per patient, sensitivity ≈ **0.33 vs 0.13**.



≤ 20 mm: **0/8 vs 3/8**; > 20 mm: 1/7 vs 2/7 (exploratory only).

RESULTS · UNCERTAINTY

A strong signal, not yet established



B – A lesion sensitivity, 95% CI

[−0.077, 0.615]

B – A mean Dice, 95% CI

[−0.112, 0.473]

Both intervals include zero: with **8 test patients** the study is underpowered. This is a statement about sample size, not about the direction of the effect.

CONCLUSION

The binding constraint is standardized, well-annotated data — not model capacity

Contributions

- Phase-aware, provenance-documented liver MRI dataset
- Leakage-free protocol with frozen thresholds
- Controlled single-variable multiphase test
- Explicit statement of the power limit

Limitations

- 8 test patients, 1 negative; 33:5 imbalance
- Single centre, retrospective, no external test
- Rigid registration only
- Only 24 of 89 lesions biopsy-proven

Future work

- Larger, powered A-vs-B cohort
- Multi-centre, multi-vendor data
- Multi-rater annotation; elastic registration
- False-positive suppression; open artifacts

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